

Secondary Examination, 2017

Mathematics

Time- $3\frac{1}{4}$ Hours

M.M- 70

GENERAL INSTRUCTIONS TO THE EXAMINEES:

1. There are in all seven questions in this question paper.
2. All the questions are compulsory.
3. In the beginning of each, the number of parts to be attempted has been clearly mentioned.
4. Marks allotted to the questions are indicated against them.
5. Start from the first question and proceed to the last one.
6. Do not waste your time over a question you cannot solve

1. Q(i) Evaluate adding of inverse $\frac{x^2 + 2}{x - 1}$

Sol.

Simplify the expression,

$$\begin{aligned}\frac{x^2 + 2}{x - 1} &= -\left(\frac{x^2 + 2}{x - 1}\right) \\ &= \frac{x^2 + 2}{-x + 1} \\ &= \frac{x^2 + 2}{1 - x}\end{aligned}$$

Hence, the value of $\frac{x^2 + 2}{1 - x}$.

- (ii) Find the median of 2,12,0,9,5,15,7,4.

Sol.

Simplify the expression,

Total terms of 8.

$n = 8$, 8 is the even number.

We know that,

$$\begin{aligned}
 \text{mean} &= \frac{1}{2} \left[\left(\frac{n}{2} \right)^{\text{th}} \text{ term} + \left(\frac{n+2}{2} \right)^{\text{th}} \text{ term} \right] \\
 &= \frac{1}{2} \left[\left(\frac{8}{2} \right)^{\text{th}} \text{ term} + \left(\frac{8+2}{2} \right)^{\text{th}} \text{ term} \right] \\
 &= \frac{1}{2} [4^{\text{th}} \text{ term} + 5^{\text{th}} \text{ term}] \\
 &= \frac{1}{2} \left[\frac{5+7}{2} \right] = \frac{12}{2} = 6
 \end{aligned}$$

Hence, the value of 6.

(iii) Evaluate $\cos 15^\circ$.

- (i) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
- (ii) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
- (iii) $\frac{1}{2\sqrt{2}}$
- (iv) $\frac{\sqrt{3}\pm 1}{2\sqrt{2}}$

Sol.

Simplify the expression,

$$\begin{aligned}\cos 15^\circ &= \cos(45^\circ - 30^\circ) \\ \cos(A - B) &= \cos A \cos B + \sin A \sin B \\ \cos(A - B) &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{\sqrt{3}+1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}+1}{2\sqrt{2}}\end{aligned}$$

Hence, the value of $\frac{\sqrt{3}+1}{2\sqrt{2}}$.

(iv) Find the gradient of line perpendicular to $x - y + 3$.

(i) -1

(ii) $-\frac{1}{2}$

(iii) $\frac{1}{2}$

(iv) 1

Sol.

Simplify the expression,

$$x - y + 3 = 0$$

$$y = mx + c$$

Comparing the equation $y = mx + c$.

$$m_1 = 1$$

Let slope is m_2

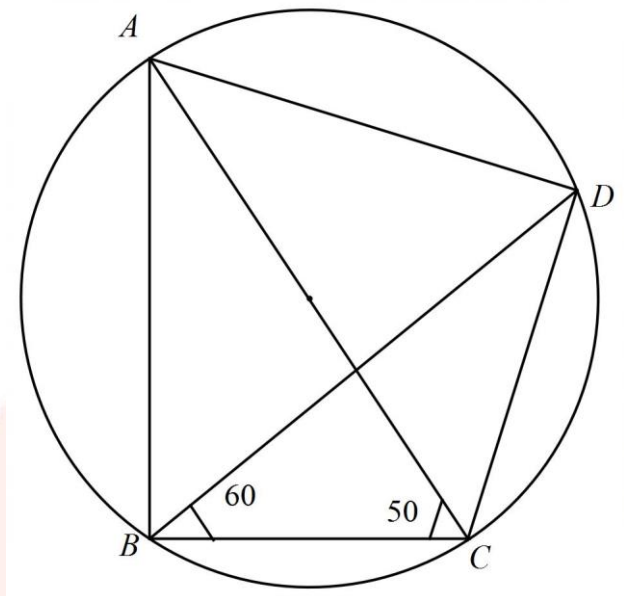
$$m_1 \times m_2 = -1$$

$$1 \times m_2 = -1$$

$$m_2 = -1$$

Hence, the value of $m_2 = -1$.

(v) $\angle ABC = 60^\circ, \angle ACB = 50^\circ$ find the angle $\angle BDC$.



Sol.

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\angle A + 60^\circ + 50^\circ = 180^\circ$$

$$\angle A = 180^\circ - 110^\circ$$

$$\angle A = 70^\circ$$

Angle of made on circumference is equal .

Therefore,

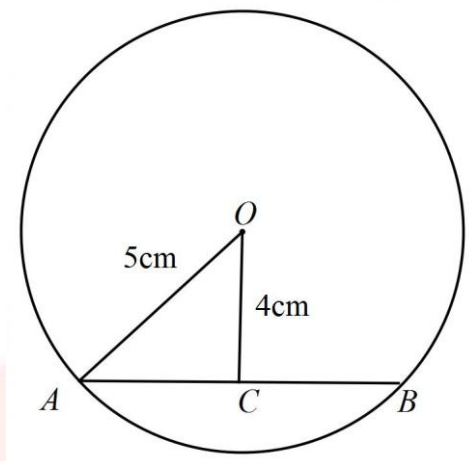
$$\angle A = 70^\circ$$

$$\angle D = 70^\circ.$$

(vi) $OA = 5\text{cm}$, $OC \perp AB$ find AB

Sol.

Simplify the expression,



s

$$AC = CB$$

In $\triangle OAC$ -

$$(AO)^2 = (OC)^2 + (AC)^2$$

$$(5)^2 = (4)^2 + (AC)^2$$

$$25 = 16 + AC^2$$

$$\Rightarrow 25 - 16 = AC^2$$

$$\Rightarrow AC^2 = 9$$

$$\Rightarrow AC = 3$$

Then,

$$AB = AC + CB$$

$$= 3 + 3$$

$$= 6\text{cm}$$

Hence, the value of $AB = 6\text{cm}$.

2(i) Find the value of x, if 4, 5, x and 9 arithmetic mean of 7.

Sol.

Simplify the expression,

$$\text{mean} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

$$7 = \frac{4 + 5 + x + 9}{4}$$

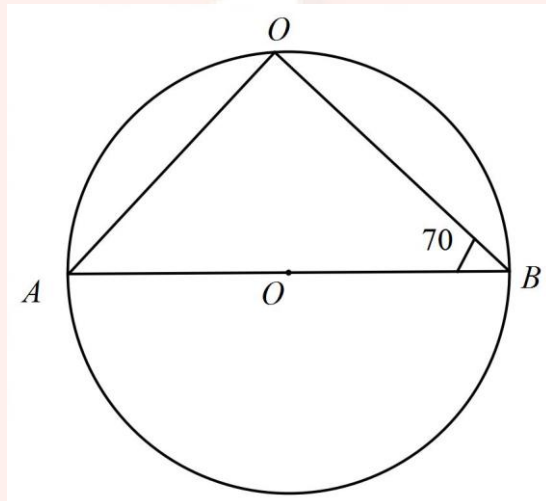
$$28 = 18 + x$$

$$x = 28 - 18 = 10$$

$$x = 10$$

Hence, the value of $x = 10$.

(ii) Given the angle $\angle ABC = 70^\circ$, then find the $\angle BAC$.



Sol.

Simplify the expression,

In $\triangle ABC$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\angle A + 70^\circ + 90^\circ = 180^\circ$$

$$\angle A + 160^\circ = 180^\circ$$

$$\angle A = 20^\circ$$

Hence, the value of $\angle A = 20^\circ$.

(iii) Given $\cos 30^\circ = \frac{\sqrt{3}}{2}$, then find the value of $\sin 15^\circ$.

Sol

Simplify the expression,

$\sin 15^\circ$,

We know that,

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$2\sin^2 \theta = 1 - \cos 2\theta$$

$$\sin \theta = \sqrt{\frac{1 - \cos 2\theta}{2}}$$

Putting the value of $\theta = 15^\circ$.

$$\sin 15^\circ = \sqrt{\frac{1 - \cos 2\theta}{2}}$$

$$\sin 15^\circ = \sqrt{\frac{1 - \cos 30^\circ}{2}}$$

$$= \sqrt{\frac{1 - \frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2 - \sqrt{3}}{2/2}}$$

$$= \sqrt{\frac{2 - \sqrt{3}}{4}}$$

$$= \frac{1}{2} \sqrt{2 - \sqrt{3}}$$

Hence, the value of $\frac{1}{2} \sqrt{2 - \sqrt{3}}$.

3. (i) $x^2 + p^2x + q = 0$ of root α, β . then find the value of $\alpha^2\beta + \alpha\beta^2$.

Sol.

Simplify the expression,

Sum of roots = (coefficient of x/coefficient of x^2).

Product of roots = (constant term/coefficient of x^2).

$$\text{Sum of roots } \alpha + \beta = -\frac{b}{a}$$

$$\text{Product of roots } \alpha \cdot \beta = \frac{c}{a}$$

$$\alpha + \beta = -\frac{p^2}{1}$$

$$\alpha + \beta = -p^2$$

$$\alpha \cdot \beta = \frac{q}{1}$$

$$\alpha \cdot \beta = q$$

Now,

$$\alpha^2 \beta + \alpha \beta^2 = \alpha \beta (\alpha + \beta)$$

$$= q \cdot (-p^2)$$

$$= -p^2 q$$

$$\alpha^2 \beta + \alpha \beta^2 = -p^2 q$$

Hence, the value of $-p^2 q$.

(ii) Mohan purchased a bicycle sales tax of Rs 1058.40, if the actual price is Rs 980 then find the rate of sales tax

Sol.

Simplify the expression,

$$M.R.P + tax \text{ account} = C.P$$

$$980 + \frac{980 \times x}{100} = 1058.40$$

$$\frac{980 \times x}{100} = 1058.40 - 980$$

$$\frac{980 \times x}{100} = 78.40$$

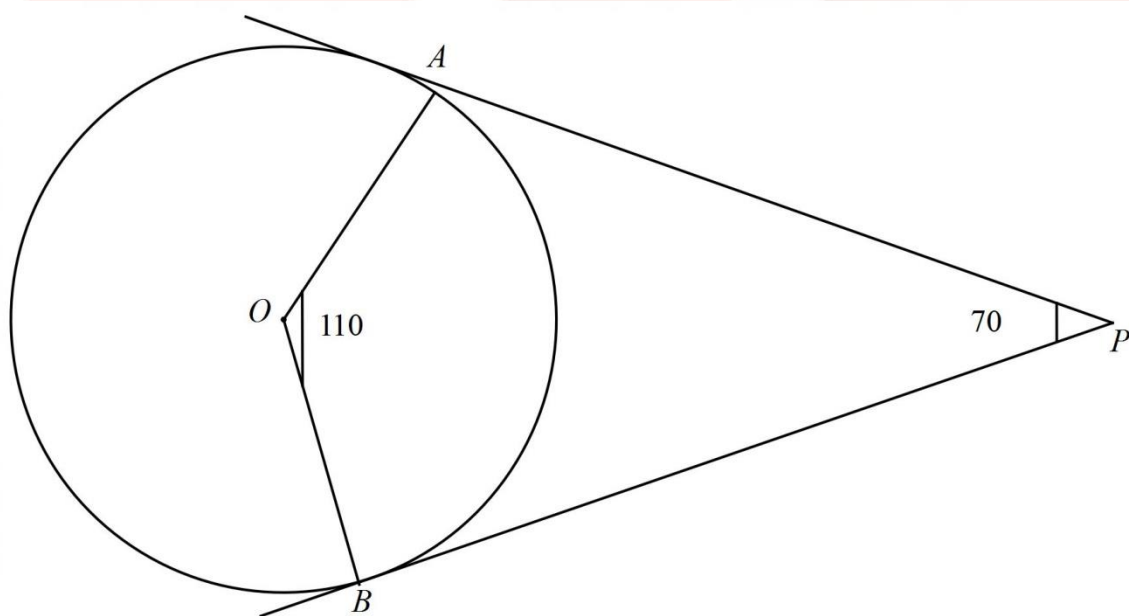
$$x = \frac{78.40 \times 100}{980}$$

$$x = 8\%$$

Hence, the value of $x = 8\%$.

(iii) Draw the circle whose radius is 2.5cm. and such that the two tangent of circle their between angle of 70° the

Sol.



Simplify the expression,

1. Draw a circle of radius 2.5cm & center O.
2. Take a point A on circle , Join OA
3. Draw a perpendicular to OA at A.
4. Draw a Radius OB, making on angle 70° with OA.
5. Draw a perpendicular to OB at B, let these perpendicular intersect at point P.

6. PA & PB are required tangents inclined at the angle of 110° .

(iv). Find the curved surface area of cylinder. If the given is cylinder of height 5cm and area of base $36\pi\text{cm}^2$

Sol

Simplify the expression,

$$h = 5\text{cm}$$

$$\pi r^2 = 36\pi$$

$$\text{curved surface area} = 2\pi rh$$

$$\pi r^2 = 36\pi$$

$$r = 6\text{cm}$$

$$\text{curved surface area} = 2\pi rh$$

$$= 2 \times \pi \times 6 \times 5$$

$$= 60\pi\text{cm}^2$$

Hence, the value of curved surface area is $60\pi\text{cm}^2$.

4(i) Two right circular cone having same height . If the radius of base of a right circular cone is half of the radius of the another cone, then find the ratio of their volume.

Sol.

Simplify the expression,

case(ii)

Radias = R

hight = H

$$v_1 = \frac{1}{3} \pi r^2 h$$

$$v_1 = \frac{1}{3} \pi r^2 h$$

case(i)

hight = H

$$\text{Radius} = \frac{R}{2}$$

$$v_1 = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi \left(\frac{R}{2} \right)^2 \cdot H$$

$$= \frac{1}{3} \pi \cdot \frac{R^2}{4} H$$

$$\frac{v_1}{v_2} = \frac{\frac{1}{3} \pi \cdot \frac{R^2}{4} H}{\frac{1}{3} \pi R^2 \cdot H} = \frac{1}{4}$$

$$v_1 : v_2 = 1 : 4$$

(ii) Find the line of perpendicular equation $\frac{x}{a} + \frac{y}{b} = 1$ such that passing through the root point.

Sol.

Simplify the expression,

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{bx + ay}{ab} = 1$$

$$bx + ay = ab \dots\dots\dots(i)$$

equation of line perpendicular to given equation.

$$ax - by = \lambda \dots\dots\dots(ii)$$

since, line passes through origin.

$$ax_0 - by_0 = \lambda$$

$$\lambda = 0$$

putting the value of λ in equation (ii)

$$ax - by = 0$$

$$ax = by$$

Hence, we get the $ax = by$.

(iii) Prove that $\cot A + \tan A = 2 \operatorname{cosec} 2A$.

Sol.

Simplify the expression,

L.H.S

$$\cot A + \tan A$$

$$\frac{\cos A}{\sin A} + \frac{\sin A}{\cos A}$$

$$= \frac{\cos^2 A + \sin^2 A}{\sin A \cdot \cos A}$$

$$= \frac{1}{\sin A \cdot \cos A}$$

Multiply by numerator and denominator in 2.

$$= \frac{2 \cdot 1}{2 \cdot \sin A \cdot \cos A}$$

We know that,

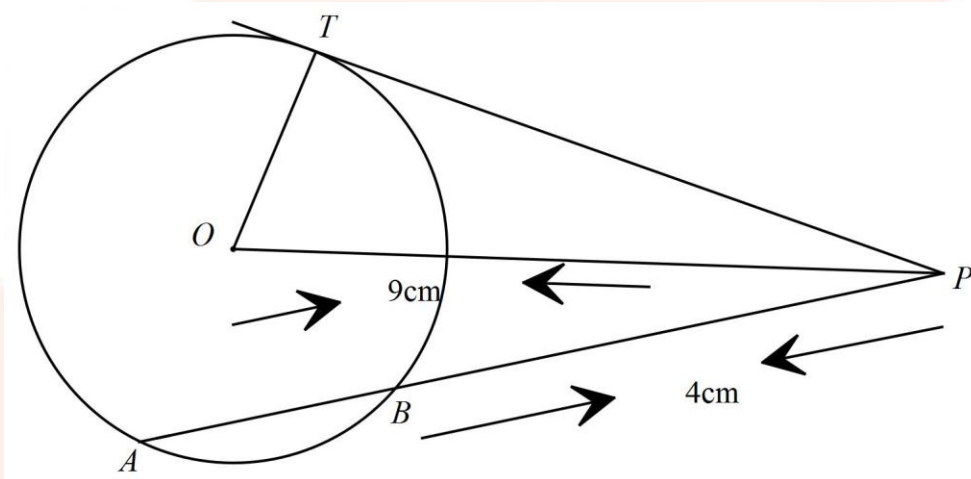
$$\sin 2A = 2 \sin A \cdot \cos A$$

$$= \frac{2}{\sin 2A} = 2 \operatorname{cosec} 2A$$

Hence, proved it.

(iv) PT is a tangent and PBA is a piercing line. Then PB=4cm, PA=9cm, PT=?.

Sol.



Simplify the expression,

$$(PT)^2 = PA \times PB$$

$$(PT)^2 = 9 \times 4$$

$$PT^2 = 36$$

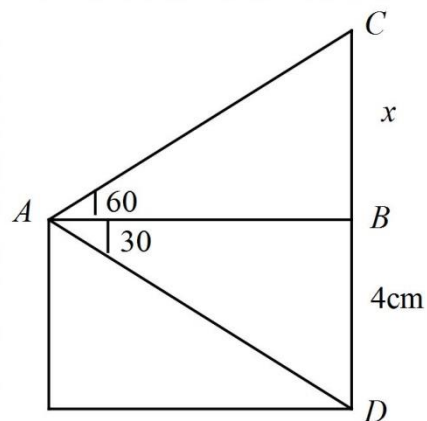
$$PT = 6\text{cm}$$

Hence, the value PT=6cm.

5.(i) A man on the deck of a ship a m above water level observe that the angle of elevation of the top of the cliff is 60° and the angle of depression of the base of cliff is 30° . Calculate the distance of the cliff from ship and height of the cliff.

Sol.

Simplify the expression,



In $\triangle ABC$ –

$$\tan 60^\circ = \frac{x}{y}$$

$$\sqrt{3} = \frac{x}{y}$$

$$y = \frac{x}{\sqrt{3}} \dots\dots(i)$$

In $\triangle ABD$ –

$$\tan 30^\circ = \frac{a}{y}$$

$$\frac{1}{\sqrt{3}} = \frac{a}{y}$$

Putting the value of y from equation (i)

$$\frac{1}{\sqrt{3}} = \frac{a}{x} \sqrt{3}$$

$$x = a\sqrt{3} \cdot \sqrt{3}$$

$$x = 3a$$

$$\text{height} = x + a$$

$$= 3a + a$$

$$= 4a$$

$$y = \frac{x}{\sqrt{3}} = \frac{3a}{\sqrt{3}}$$

$$y = \frac{\sqrt{3} \cdot \sqrt{3} \times a}{\sqrt{3}}$$

$$y = \sqrt{3}a$$

Hence, the value of $y = \sqrt{3}a$.

5(ii) Find the ratio of the height of cone and the hemi-sphere having the same base and have same volume.

Sol:

We know that,

Volume of a cone is

$$V_c = \frac{1}{3} \pi r^2 h \dots\dots (1)$$

Now, the volume of the hemi-sphere

$$V_s = \frac{2}{3} \pi r^3 \dots\dots (3)$$

Since,

$$V_c = V_s \quad (\text{Given})$$

Therefore, from eq(1) and (ii) we get

$$\frac{2}{3} \pi r^3 = \frac{1}{3} \pi r^2 h$$

$$h = 2r$$

So, the ratio of the height of the cone and the hemi-sphere is,

$$\frac{h}{r} = \frac{2}{1}$$

$$h : r = 2 : 1$$

Hence, the ratio is 2 : 1.

(iii) Find the area of the triangle. If the pick point and the equation equilateral triangle of the base is $(2,1)$ and $x + y = 2$.

Sol:

Draw perpendicular on BC,

Area of the triangle ABC,

$$= \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \times BC \times AD$$

Since, we know that length of perpendicular drawn at $x + y = 2$ from the point $(2,1)$ is

$$\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}$$

$$= \frac{2+1-1}{\sqrt{1^2 + 1^2}}$$

$$= \frac{1}{\sqrt{2}}$$

Now,

In triangle ACD,

$$\tan 30^\circ = \frac{CD}{AD}$$

$$\frac{1}{\sqrt{3}} = \frac{CD}{AD}$$

$$\frac{1}{\sqrt{3}} = \frac{CD}{\frac{1}{\sqrt{2}}}$$

$$CD = \frac{1}{\sqrt{6}}$$

$$\therefore BC = 2CD$$

$$2 \times \frac{1}{\sqrt{6}}$$

$$= \frac{2}{\sqrt{6}}$$

Therefore, the area of triangle,

$$= \frac{1}{2} \times BC \times AD$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} \times \frac{2}{\sqrt{6}}$$

$$= \frac{1}{2\sqrt{3}}$$

Hence, the area of the triangle is $\frac{1}{2\sqrt{3}}$.

- (v) If two circles having center O & O' which touches each other external point P. if a common AB drawn on AB prove that $\angle APB = 90^\circ$.

Sol.

Prove the expression,

For circle whose center is O.

CP=CA (The tangent drawn from same point)

$$\angle CAP = \angle CPA$$

$$\angle CPA = \angle CPA = \alpha$$

For another circle whose center is CB=CP (Tangent from same point).

$$\angle CBP = \angle CPB (\text{opposite angle of equal side})$$

$$\angle CBP = \angle CPB = \beta (\text{let})$$

In $\triangle APB$.

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\alpha + (\alpha + \beta) + \beta = 180^\circ$$

$$2\alpha + 2\beta = 180^\circ$$

$$\alpha + \beta = 90^\circ$$

$$\angle APB = 90^\circ [\because \alpha + \beta = \angle APB]$$

Hence, proved it.

6.a The monthly income person is Rs-50,000. He deposited Rs-35,000 prime minister fund and in which 100% discount and he denoted Rs-15,000 in a hospital on which 50% discount, then find

his saving for future per month 50,000 saving for future per month 50,000 and he deposited Rs-10,000 half-yearly life insurance also he bought national saving draft Rs-25,000. If he paid sell tax a year then the find amount of sell tax paid by his in the last month. (Rs-15,0,000 on total saving discount 100%) and education tax of 3%.

Tax of income	tax
Rs-2,00,000 to	zero
Rs-2,00,001 to 5,00,000	2,00,000 to maximum income 10%

Sol.

Simplify the expression,

$$\text{Income} = 50,000 \times 12 = 60,000$$

$$\text{Donation in prime minister relief fund} = 35,000$$

$$100\% \text{ rebate at the tax} = 35,000 .$$

$$\text{Donate in hospital} = 15,000$$

$$\begin{aligned} (50\% \text{ debate of hospital}) &= 15,000 \times 50\% \\ &= 15,000 \times \frac{50}{100} = 7,500 \end{aligned}$$

$$\text{Rest amount} = 6,00,000 - 42,500 = 5,57,500$$

$$\text{L.I.C premium} = 10,000 \times 2 = 20,000$$

$$\text{National income certificate} = 25,000$$

$$\text{Total saving} = 1,05,000$$

$$\begin{aligned} \text{Taxable amount} &= 5,57,500 - 1,05,000 \\ &= 4,52,500 \end{aligned}$$

$$(4,52,500 - 2,00,000) \times 10\%$$

$$= 2,52,000 \times 10\% = 25,250$$

$$\text{tax} = 25,200$$

$$\text{charge of education} = 25,200 \times 3\%$$

$$= 757.50$$

$$\text{Total tax} = 25250 + 757.50$$

$$= 26007.50$$

$$\text{paid tax} = 2000 \times 11 = 22,000$$

$$\text{Had to pay tax in 12th in 12th months} =$$

$$= 26007.50 - 22,000$$

$$= 4007.50$$

Hence, the final result of tax is 4007.50 .

(ii) Find the median daily wages from the following frequency distribution:

Class-interval	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	5	6	8	12	6	4

Sol.

Interval	Frequency (fi)	Mid-point (xi)	Fixi
10-15	5	12.5	62.5
15-20	6	17.5	105.0
20-25	8	22.5	180.0
25-30	12	27.5	330.0
30-35	6	32.5	195.0
35-40	4	37.5	130.0

$$\sum f_i x_i = 1022.5$$

$$f_1 = 41$$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1022.5}{41} = 24.9$$

Hence, the value of 24.9.

(iii) Evaluate the equation

$$\frac{x}{x+1} + \frac{x+1}{x} = \frac{34}{15}$$

Sol.

Simplify the equation,

$$\frac{x^2 + (x+1)^2}{x(x+1)} = \frac{34}{15}$$

$$\frac{x^2 + x^2 + 1 + 2x}{x^2 + x} = \frac{34}{15}$$

$$\frac{2x^2 + 2x + 1}{x^2 + x} = \frac{34}{15}$$

$$30x^2 + 30x + 15 = 34x^2 + 34x$$

$$4x^2 + 4x - 15 = 0$$

$$4x^2 + 10x - 6x - 15 = 0$$

$$2x(2x+5) - 3(2x+5) = 0$$

$$(2x-3)(2x+5) = 0$$

$$2x-3=0$$

$$2x=3 \Rightarrow x = \frac{3}{2}$$

$$2x+5=0$$

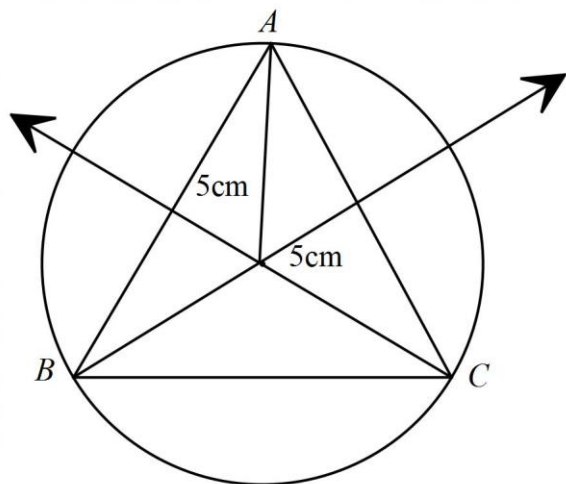
$$2x=-5$$

$$x = \frac{-5}{2}$$

Hence, the value of $x = \frac{3}{2}$ and $x = \frac{-5}{2}$.

(iv) Drawn the 5cm equilateral triangle and construction of outscribed. Give the reason.

Sol.



Step 1- Draw a line segment $BC=5\text{cm}$.

Step2-With B as center and radius 5cm .

Step3- with C as center and radius 5cm draw another arc intersecting previous arc at point A.

Step4- Join AB & AC

Step5- ABC is required equilateral triangle.

Step6- Draw perpendicular bisector of the side of triangle.

Step7- Mark the point of intersection of the perpendicular bisectors as the circumcenter of triangle.

Step8-with O as center, OA, OB, & OC are radii, now draw a circle.

Thus the circumcircle of the equilateral triangle is constructed.

7. (i) The two terms of expansion HCF and LCM $(x-5)$ and $(x-5)(x^2-7x+12)$. If one term of the expansion is $(x^2-8x+15)$ then find the second term of expansion.

Sol.

Simplify the equation,

$$\text{HCF} = x-5$$

$$\text{LCM} = (x-5)(x^2-7x+12)$$

(i) Polynomial $= (x^2-8x+15)$.

(ii) Polynomial $= ?$

$$LCM \times HCF = (i) \text{ polynomial} \times (ii) \text{ polynomial}$$

$$(x-5) \times (x^2 - 7x + 12)(x-5) = (x^2 - 8x + 15) \times (ii)$$

$$(ii) = \frac{(x-5)(x^2 - 7x + 12)(x-5)}{(x^2 - 8x + 15)}$$

$$(ii) = \frac{(x-5)^2(x^2 - 4x - 3x + 12)}{(x^2 - 8x + 15)}$$

$$(ii) = \frac{(x-5)^2[x(x-4) - 3(x-4)]}{(x^2 - 5x - 3x + 15)}$$

$$(ii) = \frac{(x-5)^2(x-4)(x-3)}{x(x-5) - 3(x-5)}$$

$$(ii) = \frac{(x-5)^2(x-4)(x-3)}{(x-5)(x-3)}$$

$$(ii) = (x-5)(x-4) = x^2 - 9x + 20$$

$$(ii) \cdot \text{polynomial} = x^2 - 9x + 20$$

Hence, the second polynomial = $x^2 - 9x + 20$.

(ii) Prove that $(\cos A + \cos B)^2 + (\sin A + \sin B)^2 = 4 \cos^2 \left(\frac{A+B}{2} \right)$.

Sol.

Prove the expression,

LHS

$$(\cos A + \cos B)^2 + (\sin A + \sin B)^2$$

We know that,

$$\cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cdot \cos \left(\frac{C-D}{2} \right)$$

$$\sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cdot \cos \left(\frac{C-D}{2} \right)$$

L.H.S

$$\begin{aligned}
 & \left[\cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cdot \cos \left(\frac{C-D}{2} \right) \right]^2 + \left[\sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cdot \cos \left(\frac{C-D}{2} \right) \right]^2 \\
 &= 4 \cos^2 \left(\frac{A+B}{2} \right) \cos^2 \left(\frac{A-B}{2} \right) + \sin^2 \left(\frac{A+B}{2} \right) \cos^2 \left(\frac{A-B}{2} \right) \\
 &= 4 \cos^2 \left(\frac{A-B}{2} \right) \left[\cos^2 \left(\frac{A+B}{2} \right) + \sin^2 \left(\frac{A+B}{2} \right) \right] \\
 &= 4 \cos^2 \left(\frac{A-B}{2} \right) \Rightarrow 1 \\
 &\Rightarrow 4 \cos^2 \left(\frac{A-B}{2} \right) = R.H.S
 \end{aligned}$$

Hence, proved it.

(iii) Prove that $\sin \alpha \cdot \sin(60 - \alpha) \cdot \sin(60 + \alpha) = \frac{1}{4} \sin 3\alpha$.

Sol.

Prove the expression,

We, know that,

$$\sin 3\alpha = 3 \sin \alpha - 4 \sin^3 \alpha$$

L.H.S

$$\sin \alpha \cdot \sin(60 - \alpha) \cdot \sin(60 + \alpha)$$

Using the formula.

$$[\sin^2 A - \sin^2 B = \sin(A+B) \cdot \sin(A-B)]$$

$$\Rightarrow \sin \alpha \left[\left(\frac{\sqrt{3}}{2} \right)^2 - \sin^2 \alpha \right]$$

$$\Rightarrow \sin \alpha \left[\frac{3}{4} - \sin^2 \alpha \right]$$

$$\Rightarrow \sin \alpha \left[\frac{3 - 4\sin^2 \alpha}{4} \right]$$

$$\Rightarrow \left[\frac{3\sin \alpha - 4\sin^3 \alpha}{4} \right]$$

$$\Rightarrow \sin 3\alpha = 3\sin \alpha - 4\sin^3 \alpha$$

$$\Rightarrow \frac{\sin 3\alpha}{4} = \frac{1}{4} \sin 3\alpha$$

$$\frac{1}{4} \sin 3\alpha = R.H.S$$

Hence, proved it.